

# A comparison study on innovation and entrepreneurship education in and out of China from a bibliometric perspective

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## Abstract

**Purpose** – In the era of the digital economy, higher demands are placed on versatile talents, and the cultivation of students with innovative and entrepreneurial abilities has become an important issue for the further development of higher education, thus leading to extensive and in-depth research by many scholars. The study summarizes the characteristics and patterns of dual-innovation education at different stages of development, hoping to provide a systematic model for the development of dual-innovation education in China and make up for the shortcomings.

**Design/methodology/approach** – This paper uses Citespace software to visualize and analyze the relevant literature in CNKI and Web of Science databases from a bibliometric perspective, focusing on quantitative analysis in terms of article trends, topic clustering, keyword co-linear networks and topic time evolution, etc., to summarize and sort out the development of innovation and entrepreneurship education research at home and abroad.

**Findings** – The study found that the external characteristics of the literature published in the field of bi-innovation education in China and abroad are slightly different, mainly in that foreign publishers are more closely connected and have formed a more stable ecosystem. In terms of research hotspots, China is still in a critical period of reforming its curriculum and teaching model, and research on the integration of specialization and creative education is in full swing, while foreign countries focus more on the cultivation of students' entrepreneurial awareness and the enhancement of individual effectiveness. In terms of cutting-edge analysis, the main research directions in China are "creative education", "new engineering", "integration of industry and education" and "rural revitalization".

**Originality/value** – Innovation and entrepreneurship education in China is still in its infancy, and most of the studies lack an overall overview and comparison of foreign studies. Based on the econometric analysis of domestic and foreign literature, this paper proposes a path for domestic innovation and entrepreneurship education reform that can make China's future education reform more effective.

**Keywords** Bibliometrics, Innovation and entrepreneurship education, Time-series analysis, Collaborative networks, Cluster analysis, Citespace

**Paper type** Research paper

## 1. Introduction

In an era of information and knowledge economy, promoting the reform of innovation and entrepreneurship education is the inevitable trend of higher education, the important support of innovation-driven development strategy and the essential requirement of building an innovative country (Wang and Feng, 2022; Wang, 2022a, b). In May 2016, the



“Outline of National Innovation-driven Development Strategy” began to be implemented. The outline is divided into three stages: China will enter the ranks of innovative countries by 2020, become the forefront of innovative countries by 2030, and become a world power in scientific and technological innovation by 2050. In 2021, China formally entered into the second stage of the “Three Steps” of “Innovation-driven Development Strategy”. Great focuses are attached to the cultivation of university students’ innovation and entrepreneurship ability, judging from the promotion of deep integration of “Production, Study and Research”, the widespread of “Internet plus” Innovation and Entrepreneurship events, and the construction of “Double First-class Universities”. The same year, the State Council issued the “Guidance on Further Supporting College Students’ Innovation and Entrepreneurship”, which put college education in the first place of reform and proposed that innovation and entrepreneurship should be carried throughout the whole process of talent training. As a key player in the education reform, universities have actively carried out many activities such as innovation and entrepreneurship simulation training courses, incubation work for students’ entrepreneurial projects, University-enterprise collaboration to build talents training bases, etc.

Although China’s innovation and entrepreneurship education reform has achieved certain results, there are still many problems in terms of policy implementation, concept inculcation and system renewal. Innovation and entrepreneurship education abroad began in the early twentieth century, nearly eighty years earlier than that in China. The stability of the innovation and entrepreneurship education ecosystem formed by them is far ahead of that of China, which has a high value for reference. The comparative analysis of the two in this paper can summarize the different research characteristics at home and abroad, take the strengths and make up for the weaknesses, and put forward suggestions and prospects for the next research in China, so as to promote the renewal of China’s innovation and entrepreneurship education system. At the same time, this study can have an impact on the relevant personnel of innovation and entrepreneurship education reform, and help the academic research, policy formulation and effect implementation of innovation and entrepreneurship education reform. The research ideas of this paper are as follows: the introduction and literature review at the beginning explain the significance, purpose and related research situation of this study; the empirical analysis in the middle is divided into “data sources”, “Research on external characteristics of literature”, “research hotspots and frontier analysis”; the final conclusion summarizes the full research, and puts forward the existing problems, solutions and impacts on relevant personnel in the development of innovation and entrepreneurship education, and explains the limitations of this paper and the prospects for future research.

## 2. Literature review

Innovation and entrepreneurship education reform has become a breakthrough to promote and deepen the comprehensive reform of higher education and has been widely concerned and recognized by scholars. Some scholars have discussed the necessity and importance of innovation and entrepreneurship education reform. [Yao and Zhang \(2023\)](#) proposed that innovation and entrepreneurship education is the breakthrough point for deepening the reform of higher education in China and an important means to cultivate high-quality innovative and entrepreneurial talents. [Wang and Cheng \(2021\)](#) pointed out that the reform of innovation and entrepreneurship education is a fundamental change of the talent training mode made by higher education to adapt to the economic and social development. [Xu \(2022\)](#) found through the questionnaire survey that the popularity and attention of innovation and entrepreneurship education are increasing in colleges and universities. [Sun \(2022\)](#) believes that promoting the construction and development of

innovation and entrepreneurship education bases has become a powerful support point and an important grip for the reform of innovation and entrepreneurship education in universities.

It is necessary for researchers and related personnel in a certain field to understand the development process of the field. To measure the development of a field, the quantitative analysis includes some fixed indicators such as the number of papers published, patents, research investments, etc. Bibliometrics is one of the more popular research methods in academia in recent years. [Vílchez-Román et al. \(2021\)](#) proved that the bibliometric method combined with data visualization is a valuable analysis method, and many scholars have applied this method in the field of education. For example, Maker Education ([Feng et al., 2022](#)), Smart Education ([Li and Wong, 2022](#); [Ma et al., 2022](#)), Urban Education ([Chen et al., 2019](#)) and TVET ([Abd Majid et al., 2022](#)). In addition to education, the research method has also been applied in other fields, such as library tech ([Kim. et al., 2021](#)), chatbots and virtual assistants ([Agarwal et al., 2022](#)), Global Digital News ([Banshal et al., 2022](#)), food ([Şahin and Yılmaz, 2022](#)), etc. Bibliometric analysis can map the knowledge of a certain field, chart its development, detect research trends and quickly help scholars to locate the frontiers and challenges in the field.

In recent years, scholars have used bibliometric methods to study the development history of innovation and entrepreneurship education in China or abroad. [Hong \(2018\)](#) made a visual analysis of the research hotspots and trends of innovation and entrepreneurship education in universities in China, using the CSSCI database and core journals as the data source, but only selected “innovation and entrepreneurship education” as a keyword for the search, so the study was not comprehensive. [Gabrielsson et al. \(2020\)](#) combed through the development of research on entrepreneurship education pedagogy through a bibliometric approach, but did not mention the concept of innovative entrepreneurship education. [Zhang et al. \(2023\)](#) selected “innovation” and “entrepreneurship” and “education” as search terms, and the article focused on the interpretation and analysis of the knowledge map.

[Du et al. \(2023\)](#) chose “innovation and entrepreneurship plus university or innovation and entrepreneurship plus college or innovation and entrepreneurship plus institution”, “innovation and entrepreneurship plus undergraduate” and “innovation and entrepreneurship plus graduate” as title search terms, the retrieval method is relatively accurate, but also no comparative analysis was done with foreign countries. [Yang et al. \(2022\)](#) used bibliometric software to sort out relevant studies on the quality evaluation of innovation and entrepreneurship education in domestic universities and put forward the shortcomings of the quality evaluation system in this field. [Nájera-Sánchez et al. \(2022\)](#) mapped the knowledge structure of entrepreneurship education through a bibliometric approach and verified that entrepreneurial intention is the best variable to predict entrepreneurial behavior and that entrepreneurship education is a way to develop entrepreneurial intention.

In addition to the bibliometric approach, there are also scholars using other methods to comb and evaluate the development of innovative entrepreneurship education. For example, [He and Liu \(2022\)](#) sorted out the development of the concept and practice of innovation and entrepreneurship education in Canada through the policies and plans of the Canadian government over the years, and this study can provide reference for the development of innovation and entrepreneurship education in China from the practical level.

These existing research results provide a reference for the development of the field of innovation and entrepreneurship education, but there are also problems such as single research method, insufficient research scope and short research period. On the basis of previous research, this paper will optimize the research ideas, deepen the research content and more systematically analyze the development process at home and abroad, and enrich the research results in this field.

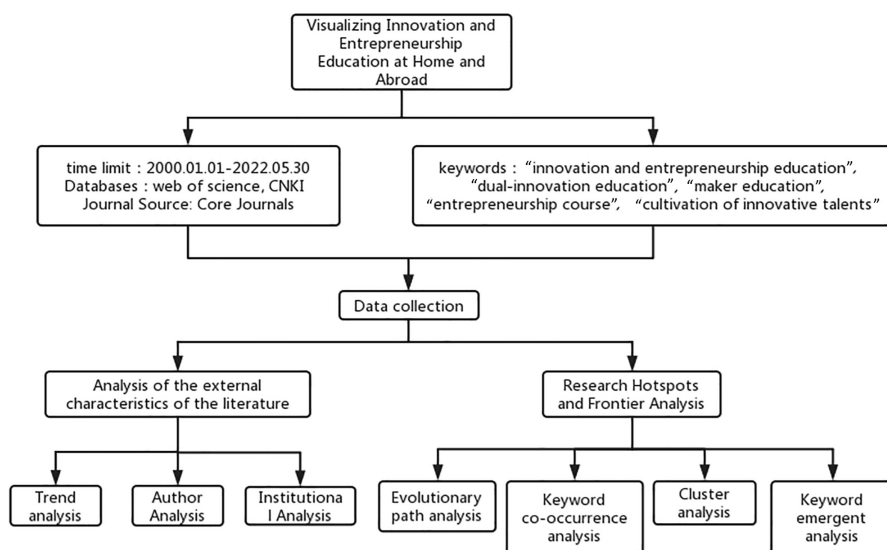
### 3. Data sources and research methods

#### 3.1 Data sources

The domestic research uses CNKI database-related documents as the source and “innovation and entrepreneurship education”, “dual-innovation education”, “maker education”, “entrepreneurship course”, “cultivation of innovative talents” as the search terms. A total of 6,440 papers in core journals from January 1, 2000, to May 30, 2022, were accurately retrieved. The foreign research used the relevant documents of the core collection in the *Web of Science* database, and “innovation and entrepreneurship education”, “dual-innovation education”, “maker education”, “entrepreneurship course”, “cultivation of innovative talents” as the search term. By removing sources such as conferences and books, and removing publications by Chinese authors, a total of 3,520 papers from 1 January 2000 to 30 May 2022 were accurately searched. As a result, all research in this paper will be based on these two types of data.

#### 3.2 Research methods

The method of this study is bibliometric, with “co-occurrence clustering” as the starting point. In the section of external characteristics analysis, we analyze the development trend of domestic and international research on dual-innovation education by drawing a line graph of article volume comparison and analyze the cooperation relationship between researchers by drawing an author/institution co-occurrence network. In the section of research hotspot and frontier analysis, the evolutionary path of dual-innovation education research is analyzed by drawing the co-citation table; the hotspot of dual-innovation education is summarized and analyzed by drawing the keyword knowledge map and topic clustering map; the frontier of dual-innovation education is detected and analyzed by drawing the keyword emergence map. The above analyses are sorted out and summarized, and different research characteristics at home and abroad can be analyzed in detail. The research framework of this paper is shown in Figure 1.



**Figure 1.**  
Research framework

**Note(s):** This figure is made by authors using WPS software

4. Analysis of external features of the literature

4.1 Time and trend analysis of publications

The chronological distribution of literature and its growth pattern can reflect its development speed, research level and scale of research in the subject area to a certain extent, and also help predict the future development trend and research dynamics (Qiu *et al.*, 2014). As early as February 1947, the opening of Harvard Business School’s MBA course “Management of the New Enterprise” implicated that entrepreneurship education for universality students began to emerge, and now the United States and other foreign developed countries have established a perfect system of entrepreneurship education for students. At the same time, however, the domestic start is late and was severely affected because of the first outbreak of epidemic in Wuhan. In the post-epidemic era, although all countries are affected by COVID-19 to certain degrees, the relevant policies released by each country witness the government’s attention to the development of dual-entrepreneurship education will increase instead of decrease. To present more intuitively the number of domestic and foreign publications in the field of innovation and entrepreneurship education and the comparison between domestic and foreign, the analysis by EXCEL is presented in the form of a line graph, see Figure 2.

4.2 Author analysis

Core authors have both broad and narrow interpretations. They are narrowly defined as someone who has an in-depth knowledge of the field and has made outstanding contributions; In a broad sense, they have relatively good writing ability and professional skills and a high number of publications (Li, 2013). In this paper, we choose a broad interpretation. According to Price’s law, the number of publications of core authors needs to be approximately greater than  $0.749 * \sqrt{n_{max}}$  ( $n_{max}$  indicates the number of publications of the author with the most publications), and it can be seen from the statistics that the maximum number of publications in this field is 14, so the number of publications greater than or equal to three is the core authors, and there are 133 core authors in this field. Table 1 shows the authors who have published more than 5 articles, totaling 13. The total number of domestic authors of publications is 845, and the authors with the first number of publications, Zhanren Wang and Zhiqiang Wang, published 14 papers in this field respectively. In conclusion, many researchers have launched studies on innovation and entrepreneurship education, but most authors have only started preliminary studies. In terms of quality, the number of relevant core authors in China is still small, and neither is the degree of cooperation

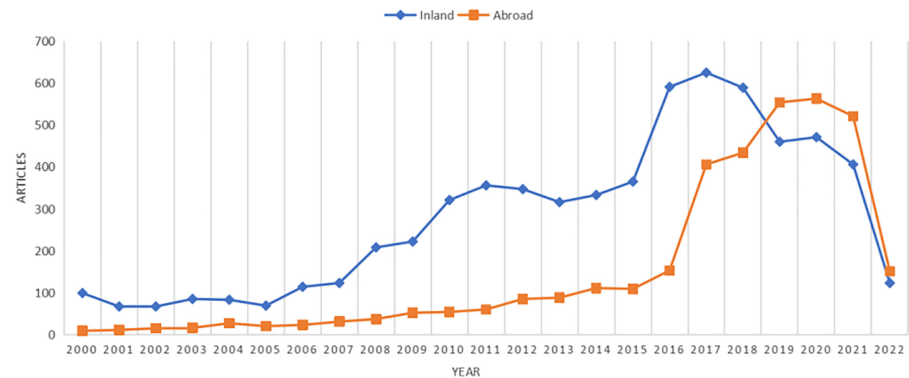


Figure 2.  
Number of domestic  
and international  
publications

Note(s): This figure is made by authors using EXCEL software

| Authors                                                                          | Count | Year | Innovation and entrepreneurship education                 |
|----------------------------------------------------------------------------------|-------|------|-----------------------------------------------------------|
| Wang Zhanren                                                                     | 14    | 2015 | <div>1815</div>                                           |
| Wang Zhiqiang                                                                    | 14    | 2016 |                                                           |
| Zhuo Zelin                                                                       | 13    | 2016 |                                                           |
| Zhang Xiongwu                                                                    | 12    | 2010 |                                                           |
| Wang Hongcai                                                                     | 11    | 2016 |                                                           |
| Editorial Board of the Journal                                                   | 10    | 2010 |                                                           |
| Zhu Changping                                                                    | 9     | 2009 |                                                           |
| Wu Aihua                                                                         | 9     | 2014 |                                                           |
| Xu Xiaozhou                                                                      | 7     | 2010 |                                                           |
| Han Fang                                                                         | 7     | 2017 |                                                           |
| Zhang Bin                                                                        | 6     | 2013 |                                                           |
| Zhang Chao                                                                       | 6     | 2017 |                                                           |
| Li Juan                                                                          | 6     | 2013 |                                                           |
| <b>Note(s):</b> This table is made by authors using EXCEL and Citespace software |       |      | <b>Table 1.</b><br>Number of articles by domestic authors |

and the depth of research. Therefore, there is still much room for improvement in the research of theoretical innovation and entrepreneurship education in China.

Figure 3 shows the correlation between the authors of the papers, which is scattered overall, but several regions show a complex net-like distribution. One team member includes Zhang Shouping, Qi Xiaoli, Zhang Shulan, Zhang Haijun, Wang Changbao, Xue Yong, etc. The members of this research group are all from Jiamusi University, and one of the co-authored papers is titled “Exploration and practice of innovation and entrepreneurship talent training model reform in local universities” (Zhang *et al.*, 2017); another group member includes Zhang Jiahua, Tian Maochun, Li Nengzhang, Wei Xueliang, Wang Wei, Peng Yanyi, Wang Haoju, Xiao Xiong, Wang Zili, Liu Li, Wu Jianyun, Chen Jianbo, Wang Xianzhong, etc., and one of the co-authored articles is titled “Exploration of innovative talent cultivation model of veterinary medicine in comprehensive



**Note(s):** This figure is made by authors using Citespace software

**Figure 3.**  
Domestic author co-occurrence mapping (partial)

universities” (Wang *et al.*, 2016), and these two articles are ranked in CNKI database. The number of citations of these two articles in the CNKI database is also high in the ranking of related topics. In addition, Wang and Zhuo Zelin, who are among the top three in terms of the number of articles published, also have a collaborative relationship so far. The emergence of the research community has made the research topic more practical and theoretically deeper, providing a partial reference for other researchers to do further research in the future. Meanwhile, the appearance of research communities has helped to create continuity, breadth and depth in research.

Similarly for the analysis of foreign authors, the related literature from Web of Science database is used as the data source, and the time span and time slices are chosen to be consistent with those in China. The data processing results indicated that the total number of people who published relevant studies abroad was 704. The author with the highest number of publications, GIUSTINA SECUNDO, published 11 articles; only 4 people published more than 5 articles, accounting for 0.57% of the authors who published; 462 authors published only one relevant article, accounting for 65.6% % of the authors. Similarly, foreign research on innovation and entrepreneurship education faces the structural contradiction of having more researchers in terms of number but not enough research in-depth. In terms of authorship collaboration, foreign countries also show most of the dispersion, a small part of the aggregation of the phenomenon, as shown in Figure 4. Table 2 shows the authors with three or more publications, 13 in total.

4.3 Institution analysis

Through the visual analysis of domestic-related research institutions, universities are the main force in the field of innovation and entrepreneurship education reform in China. However, the geographical distribution of major issuing universities is not equally distributed with Northeast and East China being more prominent. Among them, the College of Education of Zhejiang



Figure 4.  
Co-presentation map of  
foreign authors  
(partial)

Note(s): This figure is made by authors using Citespace software



| Authors | Count | Year | Innovation and entrepreneurship education<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br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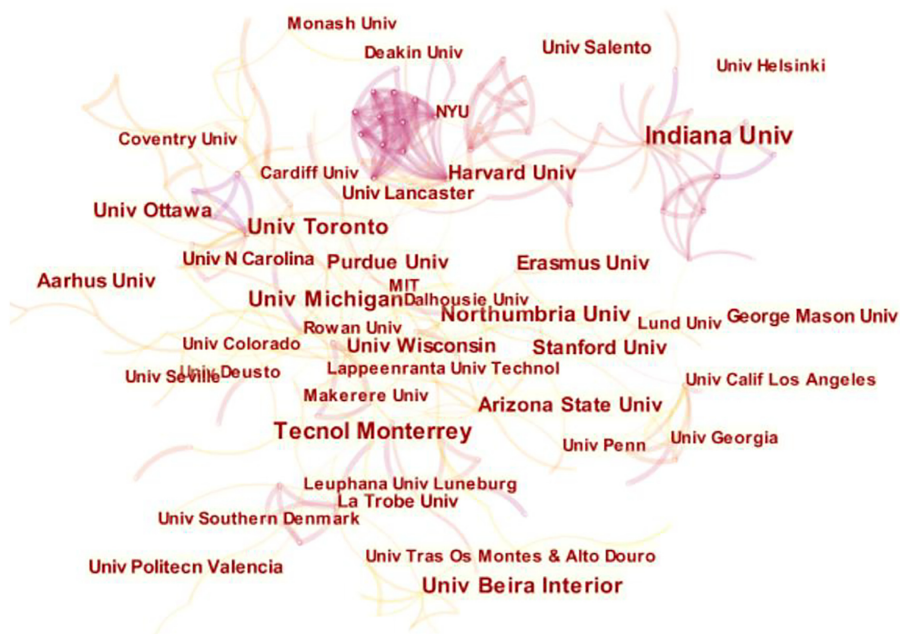
University has the most number of publications, and the Research Center for Ideological and Political Education of Northeast Normal University is the second. In addition, among the top 20 institutions in terms of the number of articles issued, as shown in [Table 3](#), teacher-training colleges and institutes of education account for 10, which shows that teacher-training colleges and institutes of education are more inclined to research innovation and entrepreneurship education. However, since there is no cooperation between the issuing institutions, the co-presentation chart is not illustrated below.

Among foreign issuing institutions, universities continue to come out on top and are the cradle of innovation and entrepreneurship education reform. Universities at home and abroad are also following the pace of the country and the needs of the times, becoming the forefront of educational development and conducting a vigorous and comprehensive theoretical exploration of “double-creation” education. [Figure 5](#) shows that there are intertwined

| Institutions                                                                         | Count | Year |
|--------------------------------------------------------------------------------------|-------|------|
| College of Education, Zhejiang University                                            | 17    | 2009 |
| Research Centre for Ideological and Political Education, Northeast Normal University | 12    | 2016 |
| Jiangsu Agricultural and Animal Husbandry Technology Vocational College              | 11    | 2015 |
| Research Centre for Higher Education Development, Xiamen University                  | 8     | 2017 |
| Zhejiang Finance Vocational College                                                  | 8     | 2016 |
| Changzhou Engineering Vocational and Technical College                               | 8     | 2015 |
| Wenzhou University Entrepreneurship Training Institute                               | 7     | 2011 |
| Xi'an Jiaotong University                                                            | 6     | 2016 |
| Jilin Agricultural University                                                        | 6     | 2016 |
| South China University of Technology                                                 | 6     | 2010 |
| Northeast Agricultural University                                                    | 6     | 2013 |
| Northeast Normal University                                                          | 6     | 2011 |
| Zhejiang Institute of Economic and Trade Technology                                  | 6     | 2012 |
| College of Education Science, Guizhou Normal University                              | 5     | 2018 |
| Yiwu Vocational and Technical College of Industry and Commerce                       | 5     | 2019 |
| East China Normal University                                                         | 5     | 2010 |
| China Academy of Educational Sciences                                                | 5     | 2013 |
| Department of Education, South West University                                       | 5     | 2017 |
| Northeastern University                                                              | 5     | 2009 |
| College of Education Science, Guizhou Normal University                              | 5     | 2018 |
| <b>Note(s):</b> This table is made by authors using EXCEL and Citespace software     |       |      |

Table 3.  
Volume of articles  
issued by domestic  
institutions





**Figure 5.**  
Co-presence mapping  
of foreign issuing  
institutions (partial)

**Note(s):** This figure is made by authors using Citespace software

and complex partnerships between foreign universities in the field of innovation and entrepreneurship education. Collaborative innovation among colleges and universities can break the limitation of ideas and geographical restrictions, which is conducive to the development of cross-disciplinary fields at the same time. By integrating the resources of innovation and entrepreneurship education, its effectiveness can be maximized. The process of collaboration in higher education requires the support of many stakeholders and involves many stakeholders, which also proves that the foreign system of dual education is very mature. Institutions with 10 or more publications are shown in [Table 4](#).

**5. Analysis of research hotspots and frontiers**

*5.1 Evolutionary path analysis*

*5.1.1 Domestic evolutionary path analysis.* In the field of innovation and entrepreneurship education, although the domestic research on it started late, the number of studies became explosive growth in the short years after the start, and the domestic-related literature screened in this paper was cited 136,844 times, with an average citation of 21.25 times (see [Table 5](#)). By analyzing the literature with high frequency, domestic innovation and entrepreneurship education research can be roughly divided into the following three research stages:

The first is the initial stage. The research in this period mainly starts from the perspective of concept interpretation and system construction and stays at the stage of system conception of how to implement innovation and entrepreneurship education. [Nan \(2001\)](#) grasped the two missions of the times: popularizing information technology education and cultivating innovative talents, and explored what information technology education is, what innovative talents are, and how to use information technology to promote the cultivation of innovative

| Institutions           | Count | Year |
|------------------------|-------|------|
| Indiana Univ           | 31    | 2000 |
| Tecnol Monterrey       | 22    | 2017 |
| Univ Beira Interior    | 21    | 2012 |
| Univ Toronto           | 20    | 2000 |
| Northumbria Univ       | 18    | 2017 |
| Univ Michigan          | 18    | 2017 |
| Purdue Univ            | 16    | 2008 |
| Arizona State Univ     | 16    | 2012 |
| Harvard Univ           | 16    | 2002 |
| Univ Wisconsin         | 15    | 2010 |
| Aarhus Univ            | 14    | 2008 |
| Stanford Univ          | 14    | 2012 |
| Erasmus Univ           | 13    | 2011 |
| Univ Ottawa            | 13    | 2019 |
| La Trobe Univ          | 12    | 2006 |
| Univ Salento           | 12    | 2017 |
| Univ Politecn Valencia | 12    | 2013 |
| Univ Valencia          | 11    | 2012 |
| Univ N Carolina        | 11    | 2013 |
| George Mason Univ      | 11    | 2006 |

**Table 4.**  
Volume of articles  
issued by foreign  
institutions

**Note(s):** This table is made by authors using EXCEL and Citespace software

talents. [Zheng and Shao \(2007\)](#) analyzed the gap between the current situation of practical teaching in colleges and universities and the cultivation of innovative talents, constructed an integrated practical teaching system in colleges and universities to cultivate innovative practical ability according to the connotation requirements of students' innovative practical ability, and proposed relevant guarantee measures for implementation. In this paper, [Qin \(2013\)](#) provided a guiding framework for the deepening of the current classroom teaching model, as well as a reference for transforming the existing classroom education concept in China, solving the problem of reducing the burden of primary and secondary school students and innovating the path of talent cultivation.

The second is the middle stage. As innovation and entrepreneurship education enters the implementation period, how to turn theory into practice and improve the innovation and entrepreneurship education system promptly becomes the focus of scholars' attention. [Yu and Hu \(2015\)](#) proposed an interdisciplinary integration model based on STEM curriculum through the elaboration of nine core features of STEM education, with a view to promoting the healthy development of STEM education practice in China. [Ma and Bai \(2015\)](#) refined the problems that existed at the early stage of innovation and entrepreneurship education practice, and introduced the "university-government-enterprise" ecological network theoretical system to provide new ideas for the study of the practice model of innovation and entrepreneurship education. [Zhu and Luo \(2015\)](#) summarized the concept and characteristics of creator education, analyzed the advantages of implementing creator education in China and put forward suggestions for future development. [Wang \(2015\)](#) proposed a "broad-spectrum" innovation and entrepreneurship education concept and model with the structure of "generalist", "embedded", "professional" and "vocational", which has important theoretical value for pursuing the concept of effective innovation and entrepreneurship education and rationalizing the relationship between innovation and entrepreneurship education and employment education. [Zhou \(2016\)](#) recognized that the key to the construction of "double first-class" universities is to promote the integration of disciplinary and professional curricula.

| Sort | Cited title                                                                                                                                             | Cited author                                                                      | Cited source                    | Publication<br>year | Cited |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------|---------------------|-------|
| 1    | STEM Education and Its Model for Interdisciplinary Integration                                                                                          | Yu Shengquan; Hu Xiang                                                            | Open Education Research         | 2015                | 1,482 |
| 2    | Maker Education: a Practical Field of ICT-Enabling Innovation Education                                                                                 | Zhu Zhiting; Sun Yanyan                                                           | China Educational Technology    | 2015                | 1,001 |
| 3    | Research and Practice Mode of China's Innovation and Entrepreneurship Education                                                                         | Ma Yongbin; Bai Zhe                                                               | Tsinghua Journal of Education   | 2015                | 761   |
| 4    | Flipped Learning: A New Paradigm for Classroom Reform                                                                                                   | Qin Weiwei                                                                        | e-Education Research            | 2013                | 654   |
| 5    | Integrating innovation and entrepreneurship education into the talent cultivation system of colleges and universities                                   | Li Jiahua; Lu Xudong                                                              | China Higher Education          | 2010                | 613   |
| 6    | On the integration of entrepreneurship education and professional education in universities                                                             | Huang Zhaoxin; Wang Zhiqiang                                                      | Educational Research            | 2013                | 586   |
| 7    | From Creativity Movement to Creativity Education: Cultivating a Culture of Creativity                                                                   | Zhu Zhiting; Luo Liang                                                            | e-Education Research            | 2015                | 574   |
| 8    | On the positioning of application-oriented universities                                                                                                 | Pan Maoyuan; Che Rushan                                                           | Journal of Higher Education     | 2009                | 573   |
| 9    | The construction and implementation of the practical teaching system in colleges and universities to cultivate innovative practical ability             | Zheng Chunlong; Shao Hongyan                                                      | China Higher Education Research | 2007                | 532   |
| 10   | On the Systematic Framework and Theoretical Value of the "University-wide" Innovation and Entrepreneurship Education                                    | Wang Zhanren                                                                      | Educational Research            | 2015                | 526   |
| 11   | Deep integration of innovation and entrepreneurship education with professional education                                                               | Liu Yan; Yan Guodong; Meng Wei; Quan Yutong; Lu Jiahui; Teng Lirong; Meng Qingfan | China University Teaching       | 2014                | 517   |
| 12   | Information Technology Education and Creative Talent Cultivation (above)                                                                                | Nan Guorong                                                                       | e-Education Research            | 2001                | 508   |
| 13   | Research on the mode of entrepreneurship education for undergraduates in China                                                                          | Mu Zhirong                                                                        | Journal of Higher Education     | 2006                | 479   |
| 14   | On Academic Breakthrough in the Construction of "Double World-class" —On the Integrative Construction of Discipline, Specialty and Course in University | Zhou Guangli                                                                      | Educational Research            | 2016                | 465   |
| 15   | Connotation, Characteristics, and Implementation Path of K-12 Learning by Making in the USA                                                             | Zheng Yanlin; Li Luyi                                                             | Open Education Research         | 2014                | 458   |

**Table 5.**  
Top 25 most frequently  
cited literature in China

(continued)

|                                                                                  |                                                                                                                                                                                                                |                                     |                                                     |                  |       | Innovation and entrepreneurship education |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------|------------------|-------|-------------------------------------------|
| Sort                                                                             | Cited title                                                                                                                                                                                                    | Cited author                        | Cited source                                        | Publication year | Cited |                                           |
| 16                                                                               | Research on Ten-year Entrepreneurship Education in Chinese Universities: Evolution, Problems and System Construction                                                                                           | Li Weiming; Li chunyan; Du Xiaohua  | Educational Research                                | 2013             | 443   | <b>1821</b>                               |
| 17                                                                               | Innovation and Entrepreneurship Education - Cultivating Trailblazers for the Cause of the New Era - Review of the Seminar on Innovation and Entrepreneurship Education of the Chinese Higher Education Society | Gao Xiaojie; Cao Shengli            | China Higher Education Research                     | 2007             | 430   |                                           |
| 18                                                                               | The inspiration of foreign innovation and entrepreneurship education model in our universities                                                                                                                 | Hu Tao; Shen Li                     | China University Teaching                           | 2013             | 424   |                                           |
| 19                                                                               | On Maker Education and Innovation Education                                                                                                                                                                    | He Kekang                           | Educational Research                                | 2016             | 416   |                                           |
| 20                                                                               | The Overall Plan of Reforming Educational Ideas on HEI Innovation Entrepreneurship Education in China                                                                                                          | Wang Zhanren                        | China Higher Education Research                     | 2015             | 414   |                                           |
| 21                                                                               | Analysis of Regional Innovation Efficiency and Spatial Discrepancy in China                                                                                                                                    | Shi Xiong; Zhao Shudong; Wu Fuxiang | The Journal of Quantitative and Technical Economics | 2009             | 412   |                                           |
| 22                                                                               | The Exploration of Maker Education Oriented to STEM Education                                                                                                                                                  | Wang Xuqing                         | China Educational Technology                        | 2,105            | 412   |                                           |
| 23                                                                               | From Verification to Creation—A Research on STEM Education Performance Models in Middle and Primary Schools                                                                                                    | Fu Qian; Liu Pengfei                | China Educational Technology                        | 2016             | 398   |                                           |
| 24                                                                               | Thoughts of Construction on Personnel Cultivating System of Innovation-entrepreneurship Education in Colleges and Universities                                                                                 | Liu Wei                             | Education Science                                   | 2011             | 382   |                                           |
| 25                                                                               | Reflection and Model Construction of Innovation and Entrepreneurship Education in Higher Education                                                                                                             | Wang Yanxin                         | China University Teaching                           | 2015             | 380   |                                           |
| <b>Note(s):</b> This table is made by authors using EXCEL and Citespace software |                                                                                                                                                                                                                |                                     |                                                     |                  |       | <b>Table 5.</b>                           |

The third is the late stage. In recent years, innovation and entrepreneurship education has entered a period of optimization and transformation, and in response to the multiple dilemmas that emerge in the mid-term implementation stage, scholars have shifted their research focus to how to promote the connotation and high-quality development of innovation and entrepreneurship education in the new period. Relying on synergy theory, [Ding et al. \(2018\)](#) proposed that the different combinations of subjects in the innovation and entrepreneurship education system combine into multiple subsystems that move independently but also influence the main system synergistically, arguing that the division of power and responsibility should be clarified and the cooperation of multiple parties should be strengthened in order to achieve the best state. [Yin et al. \(2019\)](#) build a

micro-ecological cycle system of positive interaction of college students' innovation and entrepreneurship education by building four major platforms such as project-led and entrepreneurship-raising, linking the knowledge chain and practice chain of innovation and entrepreneurship education on the basis of teacher-student communication and interaction. [An \(2020\)](#) proposed a new "three-dimensional" model of "four platforms, three teams, two paths and one goal" for university innovation and entrepreneurship education. [Zhao \(2020\)](#) reconstructed a new curriculum system that deeply integrates innovation and entrepreneurship education with professional education, drawing on the experience of foreign innovation and entrepreneurship curriculum design. [Xia \(2021\)](#) argued that the current innovation and entrepreneurship education in China has the phenomenon of putting the cart before the horse, and that the reform of practical teaching should be promoted as soon as possible to integrate the concept of innovation and entrepreneurship into daily classroom teaching, so as to achieve the reasonable allocation of resources and the optimization of the effect. [Jiang and Wang \(2021\)](#) proposed a double-cycle business innovation and entrepreneurship education system based on the background of the new liberal arts construction, and proposed specific reform measures in six aspects, including building an innovation and entrepreneurship faculty and improving the curriculum system. [Shen et al. \(2022\)](#) proposed building a quality culture of innovation and entrepreneurship under the guidance of the new development concept, and proposed specific implementation paths in four directions: exploring the characteristics of the university, strengthening students' concepts, improving long-term mechanisms and enhancing deep integration. [Yang \(2022\)](#) shifted the focus of his study to the endogenous construction of core elements such as curriculum, and took the two major meta-hypotheses of innovation and entrepreneurship curriculum construction and the internal logic it follows as the entry point to build an internal strategy to drive the integration and systematization of curriculum.

*5.1.2 Foreign evolutionary path analysis.* Since the twenty-first century, international innovation and entrepreneurship education in many foreign countries has established a relatively complete support system, and the experience from its development process can provide a reference for the development of innovation and entrepreneurship education in China. In the evolutionary path analysis of foreign-related research, the literature co-citation analysis method in CiteSpace is adopted, and the total citation frequency of related journal papers is 59,164, and the average citation of each article is 15.95 times by removing the self-citation of 50,800 times, and the literature with high citation frequency and top ranking of emergence intensity will be sorted out in detail in [Table 6](#).

At the early stage of development, studying students' entrepreneurial intentions and exploring innovative entrepreneurship education models were the focus of attention of foreign scholars. In particular, a lot of scientific experimental research has been invested in the influencing factors of students' entrepreneurial intentions. [Souitaris et al. \(2006\)](#) used a pre-test-post-test quasi-experimental design based on planning theory to examine the impact of entrepreneurship programs on the entrepreneurial attitudes and intentions of science and technology students in two universities; [Liñán and Chen \(2009\)](#) used Ajzen's theory of planned behavior to develop the Entrepreneurial Intention Questionnaire (EIQ) to measure relevant cognitive constructs, testing a sample of 519 individuals from two different regions, Spain and Taiwan, to analyze its psychometric properties and measure the applicability of the entrepreneurial intention model in two different cultural environments.

In the middle of development, some scholars began to focus on the relationship between entrepreneurship education and human capital and the impact of entrepreneurship education. [Unger et al. \(2011\)](#) integrated three decades of human capital research in entrepreneurship with a meta-analysis and found a significant but small relationship between human capital and entrepreneurial success. [Martin et al. \(2013\)](#) validated the idea that "entrepreneurship education and

|                                                                                  |                                                                                          |      |          |       |      | Innovation and entrepreneurship education                               |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------|----------|-------|------|-------------------------------------------------------------------------|
|                                                                                  | References                                                                               | Year | Strength | Begin | End  |                                                                         |
| 1                                                                                | Kolb D A, 2015, EXPERIENTIAL LEARNING, V2nd, P0                                          | 2015 | 6.86     | 2015  | 2018 | 1823                                                                    |
| 2                                                                                | Unger JM, 2011, J BUS VENTURING, V26, P341, DOI 10.1016/j.jbusvent.2009.09.004           | 2011 | 9.17     | 2012  | 2016 |                                                                         |
| 3                                                                                | Neck HM, 2011, J SMALL BUS MANAGE, V49, P55, DOI 10.1111/j.1540-627X.2010.00314.x        | 2011 | 7.09     | 2013  | 2016 |                                                                         |
| 4                                                                                | Oosterbeek H, 2010, EUR ECON REV, V54, P442, DOI 10.1016/j.euroecorev.2009.08.002        | 2010 | 6.35     | 2013  | 2015 |                                                                         |
| 5                                                                                | Morris MH, 2013, J SMALL BUS MANAGE, V51, P352, DOI 10.1111/jsbm.12023                   | 2013 | 8.1      | 2014  | 2018 |                                                                         |
| 6                                                                                | Martin BC, 2013, J BUS VENTURING, V28, P211, DOI 10.1016/j.jbusvent.2012.03.002          | 2013 | 15.02    | 2015  | 2018 |                                                                         |
| 7                                                                                | Perkmann M, 2013, RES POLICY, V42, P423, DOI 10.1016/j.respol.2012.09.007                | 2013 | 12.55    | 2015  | 2018 |                                                                         |
| 8                                                                                | Abreu M, 2013, RES POLICY, V42, P408, DOI 10.1016/j.respol.2012.10.005                   | 2013 | 8.45     | 2016  | 2018 |                                                                         |
| 9                                                                                | Shinnar RS, 2012, ENTREP THEORY PRACT, V36, P465, DOI 10.1111/j.1540-6520.2012.00509.x   | 2012 | 7.46     | 2016  | 2017 |                                                                         |
| 10                                                                               | Schlaegel C, 2014, ENTREP THEORY PRACT, V38, P291, DOI 10.1111/etap.12087                | 2014 | 6.78     | 2016  | 2019 |                                                                         |
| 11                                                                               | Fayolle A, 2013, ENTREP REGION DEV, V25, P692, DOI 10.1080/08985626.2013.821318          | 2013 | 11.13    | 2017  | 2018 |                                                                         |
| 12                                                                               | Halverson ER, 2014, HARVARD EDUC REV, V84, P495, DOI 10.17763/haer.84.4.34j1g68140382063 | 2014 | 9.26     | 2017  | 2019 |                                                                         |
| 13                                                                               | Sanchez JC, 2013, J SMALL BUS MANAGE, V51, P447, DOI 10.1111/jsbm.12025                  | 2013 | 7.82     | 2017  | 2018 |                                                                         |
| 14                                                                               | Rideout EC, 2013, J SMALL BUS MANAGE, V51, P329, DOI 10.1111/jsbm.12021                  | 2013 | 7.25     | 2017  | 2018 |                                                                         |
| 15                                                                               | Sheridan KM, 2014, HARVARD EDUC REV, V84, P505, DOI 10.17763/haer.84.4.brr34733723j648u  | 2014 | 6.92     | 2017  | 2019 |                                                                         |
| 16                                                                               | Audretsch DB, 2014, J TECHNOL TRANSFER, V39, P313, DOI 10.1007/s10961-012-9288-1         | 2014 | 6.37     | 2017  | 2019 |                                                                         |
| 17                                                                               | Bae TJ, 2014, ENTREP THEORY PRACT, V38, P217, DOI 10.1111/etap.12095                     | 2014 | 15.24    | 2018  | 2019 |                                                                         |
| 18                                                                               | Fayolle A, 2015, J SMALL BUS MANAGE, V53, P75, DOI 10.1111/jsbm.12065                    | 2015 | 7.26     | 2019  | 2020 |                                                                         |
| 19                                                                               | Rauch A, 2015, ACAD MANAG LEARN EDU, V14, P187, DOI 10.5465/amle.2012.0293               | 2015 | 6.22     | 2019  | 2020 |                                                                         |
| 20                                                                               | Bacigalupo M, 2016, ENTRECOMP ENTREPRENE, V0, P0                                         | 2016 | 5.9      | 2019  | 2022 |                                                                         |
| 21                                                                               | Walter SG, 2016, J BUS VENTURING, V31, P216, DOI 10.1016/j.jbusvent.2015.10.003          | 2016 | 5.9      | 2019  | 2022 |                                                                         |
| 22                                                                               | Nabi G, 2017, ACAD MANAG LEARN EDU, V16, P277, DOI 10.5465/amle.2015.0026                | 2017 | 16.47    | 2020  | 2022 |                                                                         |
| 23                                                                               | Wright M, 2017, J TECHNOL TRANSFER, V42, P909, DOI 10.1007/s10961-017-9558-z             | 2017 | 6.89     | 2020  | 2022 |                                                                         |
| 24                                                                               | Neck HM, 2018, ENTREPRENEURSHIP ED, V1, P8, DOI 10.1177/2515127417737286                 | 2018 | 6.05     | 2020  | 2022 |                                                                         |
| 25                                                                               | Bergmann H, 2016, SMALL BUS ECON, V47, P53, DOI 10.1007/s11187-016-9700-6                | 2016 | 5.77     | 2020  | 2022 |                                                                         |
| <b>Note(s):</b> This table is made by authors using EXCEL and Citespace software |                                                                                          |      |          |       |      | <b>Table 6.</b><br>Top 25 references with the strongest citation bursts |

training (EET) can be effective in promoting human capital formation” This view is validated and future research is recommended to examine potential moderators. Oosterbeek *et al.* (2010) analyzed the impact of a leading entrepreneurship education program on entrepreneurial skills and motivation among college students. Rauch and Hulsink (2015) conducted a research study on the

impact of entrepreneurship education on entrepreneurial behavior, which showed that students who participated in entrepreneurship education showed an increase in attitudes and perceived behavioral control both improved in terms of attitudes and perceived behavioral control, and that entrepreneurial intentions are higher after entrepreneurship education. [Nabi et al. \(2017\)](#) systematically reviewed empirical evidence on the impact of entrepreneurship education (EE) in higher education on a range of entrepreneurial outcomes using a pedagogical model framework, which found that research on the impact of EE continues to focus on short-term and subjective outcome measures.

In the mid-speed and high-speed development period, with the above basic conditions in place, scholars have focused on the importance of building a favorable external social environment to promote the development of innovation and entrepreneurship education. [Bergmann et al. \(2016\)](#) explained student entrepreneurship from a contextual perspective and examines the relevance of university and regional environments on university students' entrepreneurship, pointing out that personal and environmental determinants influence students' entrepreneurial tendencies. [Ling and Dan \(2019\)](#) explored the reform of the "three combinations + three steps" teaching model in the foreign trade transportation and insurance course as an example in the perspective of innovation and entrepreneurship education and explore the construction of innovation and entrepreneurship education system based on the university perspective. [Arcega \(2021\)](#) grounded in universities and based on the business incubator model, the study identifies the practices and challenges that students experience in the business incubation services provided by the college.

## 5.2 Analysis of research hotspots

**5.2.1 Keyword co-occurrence network mapping analysis.** Keywords are the extraction of the core content of an article. 3–5 keywords usually exist in an article, covering the research topic, methods, objects and purposes of the article. Through the analysis of keywords, the focus and trends of current research can be derived. In this round, a total of 812 keywords from 6,440 Chinese documents were extracted. In the keyword co-occurrence map, the node represents a keyword, the linkage between the nodes represents the association between the nodes, and the density reflects the closeness of the linkage between the keywords. The number of nodes is 812, the number of links is 1,690 and the density is 0.0051.

To be able to display the popular keywords more accurately, the top 30 keywords are arranged in a table according to frequency, i.e. the number of keyword occurrences and the network connections between each keyword reflected by centrality, as shown in [Table 7](#). Excluding the keywords "innovation and entrepreneurship" and "entrepreneurship education", which are highly related to the index word "innovation and entrepreneurship education", other keywords that appear more frequently include "higher vocational institutions", "maker education", "college students", "universities", "talent training", "professional education", "curriculum system", "teaching reform", etc. It can be seen that the reform of innovation and entrepreneurship education is closely related to higher vocational colleges and institutions of higher education, which are supposed to carry out a comprehensive reform of their teaching, curriculum and teachers system, and to effectively integrate professional education with innovation and entrepreneurship education.

Analyzing each keyword according to the year in which it first appeared, it is easy to find that the reform of China's innovation and entrepreneurship education is a gradual process, with a gradual transition from higher vocational institutions to universities and from a widespread propaganda call to the proposal of specific reform plans, indicating China's determination to develop dual-innovation education. Besides, recent research has focused on keywords such as "new engineering", "civic education", "integration of specialization and creativity", "three-wide education", "competition for learning", "talent revitalization".

Using the same method and steps, keyword co-occurrence analysis was performed in this paper for the English literature section. The number of nodes in the map is 423, the number of



| Keywords                        | Count | Centrality | Year | Keywords                       | Count | Centrality | Year |
|---------------------------------|-------|------------|------|--------------------------------|-------|------------|------|
| Innovative Talent               | 599   | 0.25       | 2000 | Innovative Education           | 98    | 0.11       | 2000 |
| Entrepreneurship Education      | 482   | 0.11       | 2001 | Higher Education               | 95    | 0.06       | 2002 |
| Innovation and Entrepreneurship | 460   | 0.12       | 2009 | Experimental Teaching          | 80    | 0.03       | 2005 |
| Talent Cultivation              | 407   | 0.16       | 2002 | Entrepreneurship               | 70    | 0.03       | 2004 |
| Higher Education                | 279   | 0.12       | 2002 | Professional Education         | 66    | 0.03       | 2009 |
| Creativity Education            | 274   | 0.06       | 2014 | Collaborative Innovation       | 62    | 0.03       | 2012 |
| Higher Education Institutions   | 263   | 0.11       | 2007 | Local Universities             | 61    | 0.04       | 2007 |
| University Students             | 220   | 0.08       | 2004 | Entrepreneurship courses       | 59    | 0.06       | 2006 |
| Cultivation Model               | 196   | 0.06       | 2003 | Creative Space                 | 55    | 0.02       | 2015 |
| Innovation                      | 193   | 0.06       | 2002 | Entrepreneurship               | 55    | 0.04       | 2005 |
| Teaching Reform                 | 128   | 0.08       | 2005 | School-Enterprise Cooperation  | 51    | 0.05       | 2007 |
| Innovation Ability              | 122   | 0.08       | 2003 | Entrepreneurship               | 48    | 0          | 2015 |
| Cultivation                     | 110   | 0.1        | 2001 | Collaborative Education        | 45    | 0.02       | 2015 |
| Practical Teaching              | 103   | 0.05       | 2005 | Industry-Education Integration | 45    | 0.01       | 2015 |
| Curriculum                      | 99    | 0.05       | 2006 | New Engineering                | 43    | 0          | 2017 |

**Note(s):** This table is made by authors using EXCEL and Citespace software

**Table 7.**  
Frequency and centrality of domestic keywords

connections is 2,773 and the density is 0.0311. The high-frequency keywords are shown in Table 8. Similarly, some keywords such as “innovation”, “education” and “entrepreneurship” which are highly similar to the index word “innovation and entrepreneurship education” are excluded. It can be found that foreign keywords with high frequency include “performance”, “impact”, “knowledge”, “student”, “growth” and “model”, which are different from those in China and do not overemphasize the aspects of school and teaching reform. On the contrary, they mentioned more about the development of “firm”, “industry”, “self-efficiency”,

| Keywords         | Count | Centrality | Year | Keywords         | Count | Centrality | Year |
|------------------|-------|------------|------|------------------|-------|------------|------|
| Education        | 497   | 0.04       | 2004 | University       | 125   | 0.02       | 2005 |
| Innovation       | 477   | 0.03       | 2000 | Perspective      | 99    | 0.01       | 2011 |
| Impact           | 379   | 0.05       | 2008 | Enterprise       | 98    | 0.03       | 2002 |
| Performance      | 314   | 0.04       | 2005 | Determinant      | 96    | 0.04       | 2002 |
| Entrepreneurship | 261   | 0.04       | 2005 | Gender           | 95    | 0.02       | 2007 |
| Model            | 237   | 0.02       | 2008 | Behavior         | 88    | 0.04       | 2006 |
| Knowledge        | 229   | 0.04       | 2002 | Future           | 88    | 0.01       | 2015 |
| Student          | 203   | 0.02       | 2009 | Technology       | 87    | 0.04       | 2004 |
| Intention        | 170   | 0          | 2016 | Higher education | 87    | 0.01       | 2014 |
| Business         | 158   | 0.04       | 2007 | Policy           | 84    | 0.03       | 2011 |
| Science          | 158   | 0.04       | 2000 | Attitude         | 84    | 0.02       | 2003 |
| Self-efficacy    | 154   | 0.03       | 2007 | Perception       | 84    | 0.02       | 2009 |
| Growth           | 150   | 0.06       | 2002 | Framework        | 82    | 0.03       | 2015 |
| Management       | 149   | 0.02       | 2008 | Challenge        | 81    | 0.05       | 2007 |
| Firm             | 128   | 0.07       | 2001 | Strategy         | 78    | 0.04       | 2001 |

**Note(s):** This table is made by authors using EXCEL and Citespace software

**Table 8.**  
Frequency and centrality of foreign keywords

“technology” and “network”, and emphasized the role of corporate and social innovation consciousness in the cultivation of College Students’ innovation and entrepreneurship ability.

*5.2.2 Clustering analysis.* In the process of clustering analysis of keywords, you can see the topics associated with the keywords respectively, the smaller the serial number marked in front of the topic cluster, the more number of associated keywords.

In Table 9, “size” represents capacity/scale (how many keywords are included), “Silhouette” represents the degree of aggregation (the closer it is to 1, the higher the degree of aggregation), and “mean (year)” represents the average year of occurrence of this cluster. The last three columns are the keywords contained in each cluster under different algorithms. Professor Chen Chaomei’s log likelihood column is the best.

Firstly, the domestic research literature was analyzed by keyword clustering, and a total of 14 topics were formed, including #0 higher vocational colleges, #1 universities, #2 innovative talents, #3 teaching reform, #4 training mode, #5 entrepreneurship education, #6 practical teaching, #7 innovation and entrepreneurship, #8 maker education, #9 talent training, #10 entrepreneurship courses, #11 Ministry of education, #12 employment, #13 the United States, as shown in Table 9. These words are highly coincident with the high-frequency keywords in the keyword co-occurrence network map, which shows that the clustering reliability is high. Through the analysis of the cluster diagram. Through analysis,

| Cluster id | Size | Silhouette | Mean(Year) | Top terms (log likelihood ratio, <i>p</i> -level)                                                                                                        |
|------------|------|------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| #0         | 74   | 0.835      | 2013       | Higher education institutions; Innovation and entrepreneurship education; Professional education; Integration; Path                                      |
| #1         | 71   | 0.826      | 2010       | Higher education institutions; Cultivation; University students; Countermeasures; Strategies                                                             |
| #2         | 68   | 0.746      | 2010       | Innovative talents; Cultivation system; Cultivation mechanism; Graduate students; Entrepreneurship education                                             |
| #3         | 63   | 0.842      | 2010       | Teaching reform; Experimental teaching; Innovation; Innovation ability; Higher education                                                                 |
| #4         | 61   | 0.657      | 2012       | Cultivation mode; Curriculum system; Higher education; Ecosystem; Practice                                                                               |
| #5         | 58   | 0.812      | 2008       | Entrepreneurship education; Innovation education; Entrepreneurial ability; Innovation spirit; Innovation awareness                                       |
| #6         | 54   | 0.667      | 2013       | Practical teaching; Collaborative innovation; Engineering education; Technological innovation; Influencing factors                                       |
| #7         | 44   | 0.853      | 2017       | Innovation and entrepreneurship; Civic education; Innovative talents; Rural revitalization; Higher education students                                    |
| #8         | 44   | 0.754      | 2016       | Creativity education; Creativity space; Creators; Core literacy; Creativity movement                                                                     |
| #9         | 36   | 0.761      | 2010       | Talent cultivation; Artificial intelligence; Subject construction; Big data; Entrepreneurship education                                                  |
| #10        | 33   | 0.904      | 2012       | Entrepreneurship curriculum; Entrepreneurship practice; Food majors; Entrepreneurship teaching; Innovation and entrepreneurship training for food majors |
| #11        | 27   | 0.914      | 2011       | Ministry of Education; Top Students; Basic Subjects; Cultivation Experiment; Top Program                                                                 |
| #12        | 26   | 0.836      | 2013       | Employment; Entrepreneurship; Industry-education integration; Teaching model; Science-education integration                                              |
| #13        | 23   | 0.83       | 2011       | United States; Undergraduate; Undergraduate education; Personalization; Community college                                                                |

**Table 9.**  
Domestic keyword  
clustering

**Note(s):** This table is made by authors using EXCEL and Citespace software

it is found that the research focuses on domestic innovation and entrepreneurship education in the following aspects.

- (1) Reform participants (#0, #1, #11): the main initiator of innovation and entrepreneurship education reform is the Ministry of education, of which the top-notch plan initiated by the Ministry of education has the highest degree of relevance. Through the establishment of a top-notch innovation talent training reform pilot, a new mechanism for talent training is explored. With the rapid growth of the national demand for professional talents, innovation and entrepreneurship education reform has gradually expanded from colleges and universities to higher vocational colleges. Higher vocational colleges urgently need to cultivate and improve students' innovation and entrepreneurial spirit and ability. However, due to many problems in the teaching strength and management system of Higher Vocational colleges, the reform has been pushed forward with twists and turns.
- (2) Direction and process of reform (#3, #4, #6, #10): as China's ideological and political education and innovation and entrepreneurship education have great similarities in political and educational attributes, the effective integration of the two is of great value to promote the development of mass entrepreneurship and innovation education, which requires comprehensive reform of the current curriculum system, teaching mode and teaching staff of Higher Vocational Colleges and universities.
- (3) Reform objectives (#2, #12): the highest goal of innovation and entrepreneurship education reform is to provide innovation and entrepreneurship talents for the country, which is an urgent need for China to speed up the implementation of an innovation-driven development strategy. At the same time, it is of great significance to promote higher quality entrepreneurship and employment of college graduates. The focus of the reform is not to impart textbook knowledge, but to cultivate students' innovative spirit, entrepreneurial awareness, and innovative entrepreneurial ability, and strive to create a new force for mass entrepreneurship and innovation.
- (4) The concept of "hot spots of reform" (#8): maker education was first put forward in foreign countries, aiming to cultivate students' innovation and entrepreneurship awareness and action ability by carrying out entrepreneurship projects. At present, a relatively complete system has been established in some colleges and universities in China's first tier cities such as Beijing, Shanghai and Shenzhen. Other regions have also entered the stage of popularization and development, and maker education has achieved preliminary development in China. However, there are also some problems in maker education in China, including too substantial form, imperfect maker space and equipment, lack of innovative teachers, etc. (Zhou, 2021).

Based on the cluster analysis of the cited literature of foreign-related research, 12 topics are shown in Table 10, including #0 entrepreneurship education, #1 entrepreneurial intent, #2 nation culture, #3 university students, #4 entrepreneurial universities, #5 electronic textile, #6 life course pathway, #8 digital character. Issues, #9 community context, #10 industry evolution #11 enterprise capability, #12 knowledge creation, #15 predictive capability, #19 exploration study, #24 higher-order structural model, #32 enterprise collector, etc. Through analysis, it is found that the research focuses on foreign innovation and entrepreneurship education in the following aspects.

- (1) Reform participants (#3 college students, #4 entrepreneurial universities, #32 entrepreneurial lecturers): in the late twentieth century, some American research universities used their knowledge and technology innovation achievements to attract external funds to develop new industries, accelerate the transformation of research achievements, serve industrial and social development, and transform universities from

**Table 10.**  
Clustering of foreign  
cited articles

| Cluster ID | Size | Silhouette | Mean (Year) | Top terms (log-likelihood ratio, <i>p</i> -level)                                                                                                  |
|------------|------|------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| #0         | 122  | 0.86       | 2016        | Entrepreneurship education; entrepreneurial attitude; literature review; post-program research design; entrepreneurship education center           |
| #1         | 84   | 0.872      | 2016        | Entrepreneurial intention; entrepreneurial university; entrepreneurial ecosystem; university spin-off; entrepreneurship education                  |
| #2         | 82   | 0.881      | 2012        | National culture; entrepreneurial ladder; university internationalization pay; lean startup methodology; experiential learning processes           |
| #3         | 81   | 0.851      | 2014        | Entrepreneurial intention; university student; social norm; undergraduate student; engineering student                                             |
| #4         | 68   | 0.879      | 2014        | Entrepreneurial intention; entrepreneurial universities; knowledge-based society; executive MBA; moderating effect                                 |
| #5         | 45   | 1          | 2015        | Entrepreneurial intention; electronic textile; science education; electrifying engagement; middle school science class                             |
| #6         | 45   | 0.956      | 2010        | Life-course pathway; business start-up; cultural capital; socio-cultural environment; community culture motivation self-efficacy                   |
| #8         | 31   | 0.968      | 2018        | Demographic characteristics; international student; capital theory perspective; understanding digital entrepreneurial intention; contextual factor |
| #9         | 25   | 0.975      | 2006        | Community context; diagnostic framework; teachability dilemma; developmental outcome; entrepreneurial process                                      |
| #10        | 22   | 0.969      | 2010        | Industry evolution; institutional logic migration; emerging economies; changing university business mode; institutional logic                      |
| #11        | 22   | 0.995      | 2016        | Entrepreneurial capability; empirical approach; Asean-05 emerging economies; Asean-5 economies; empirical investigation                            |
| #12        | 21   | 0.974      | 2010        | Knowledge creation; SMEs sector; emotional competence; outdoor training; integrated model                                                          |
| #15        | 11   | 0.986      | 2010        | Predictive capability; business plan; evaluation; business success; project                                                                        |
| #19        | 10   | 0.99       | 2006        | Exploratory study; UK university; principal investigator role; of-concept processes; open innovation                                               |
| #24        | 7    | 0.982      | 2012        | Higher-order structural model; innovation entrepreneurship; restaurant performance; success; driver                                                |
| #32        | 6    | 0.996      | 2015        | Entrepreneurial lecturer; paid-employment intention; conceptual framework; Germany Kenya; using nominal group technique                            |

**Note(s):** This table is made by authors using EXCEL and Citespace software

- a secondary social support institution to a driving force for economic and social development Innovative research universities are called “Entrepreneurial University”, i.e. entrepreneurial universities. These universities are also the main force of innovation and entrepreneurship education reform in foreign countries. At the same time, foreign countries pay special attention to the training of innovation and entrepreneurship teachers. Such teachers will be regularly trained and the assessment system will be reset.
- (2) Reform background (#2 national culture, #6 life history, #9 community background, #10 industry evolution): the “integration of industry and education” ecosystem in foreign countries is relatively mature, and good interaction and cooperation relationship have been formed between universities, enterprises and society. The role of universities in entrepreneurial society has been expanded to focus on improving Entrepreneurial Capital and promoting prosperity in Entrepreneurial Society (Audretsch, 2014). Some scholars took Mexico as an example to explain in-depth the role of higher education in promoting the synergy of participants in the innovation ecosystem and confirmed that a good school enterprise ecosystem is conducive to the rapid growth of the social economy.

- (3) Research contents and methods (#1 entrepreneurship education, #2 entrepreneurship intention, #19 exploratory research): many foreign countries, such as France, attach great importance to “entrepreneurship education” in the concept of innovation and entrepreneurship education, and cultivate students’ basic moral cultivation and personal quality to become entrepreneurs. They will research students’ entrepreneurial intentions. [Martin et al. \(2013\)](#) and others used quantitative review for the first time to confirm that innovation and entrepreneurship education is helpful to cultivate more and better entrepreneurs. In China, there may be some misunderstanding about its training concept, which highly correlates the results of innovation and entrepreneurship education with students’ participation in competitions, and excessively pursues surface data.
- (4) Training ability (#11 entrepreneurial ability, #15 predictive ability): the focus of innovation and entrepreneurship education reform in China is on universities and enterprises, while in foreign countries, the focus is on individual students, emphasizing the improvement of individual efficiency. Teachers in many foreign schools have participated in special innovation and entrepreneurship education training and formulated different training programs for students at different stages ([Wang, 2021a, b](#)). They focus on cultivating students’ prediction ability. When students prepare a business plan, they need to predict the prospect of the project (strategic development direction, profitability, etc.).

### 5.3 Analysis of research trends and frontiers

Emergent analysis for keywords can be used to detect the frontiers of the field. [Table 11](#) shows the process of keyword emergence in the field of domestic innovation and

| Keywords                                  | Strength | Begin | End  | 2000–2022 |
|-------------------------------------------|----------|-------|------|-----------|
| Innovative talents                        | 67.31    | 2000  | 2013 |           |
| Innovative education                      | 7.35     | 2000  | 2010 |           |
| Cultivation                               | 18.17    | 2001  | 2013 |           |
| Quality education                         | 6.73     | 2003  | 2012 |           |
| Experimental teaching                     | 8.78     | 2008  | 2010 |           |
| Innovation                                | 8.47     | 2010  | 2011 |           |
| Innovation capability                     | 8.72     | 2011  | 2012 |           |
| Collaborative innovation                  | 9.42     | 2013  | 2015 |           |
| Maker education                           | 43.06    | 2016  | 2019 |           |
| Creators                                  | 12.67    | 2016  | 2018 |           |
| Creators space                            | 8.96     | 2016  | 2018 |           |
| Internet +                                | 8.02     | 2016  | 2018 |           |
| Innovation and entrepreneurship           | 32.87    | 2017  | 2020 |           |
| New Engineering                           | 15.77    | 2018  | 2022 |           |
| Higher Education Institutions             | 11.82    | 2018  | 2019 |           |
| Innovation and Entrepreneurship Education | 10.79    | 2018  | 2022 |           |
| Integration                               | 7.18     | 2018  | 2022 |           |
| New Era                                   | 6.76     | 2018  | 2022 |           |
| Ideological and Political Education       | 16.11    | 2019  | 2022 |           |
| Integration of industry and education     | 14.22    | 2019  | 2022 |           |
| Collaborative Education                   | 10.03    | 2019  | 2022 |           |
| Rural revitalization                      | 7.79     | 2019  | 2022 |           |
| Food specialties                          | 8.41     | 2020  | 2022 |           |
| Labor education                           | 8.41     | 2020  | 2022 |           |
| Chemical industry                         | 6.88     | 2020  | 2022 |           |

**Table 11.**  
Top 25 keywords with the strongest citation bursts

**Note(s):** This table is made by authors using EXCEL and Citespace software

entrepreneurship education. To better reflect the concept of frontier prediction, this paper will only analyze the emergent words after 2016.

- (1) “Creativity Education”, “Creators”, “Creators Space”: Keywords related to creators mainly broke out in 2016–2018, this has been analyzed in detail in the domestic keyword clustering section.
- (2) “New Engineering”, “New Era”, “Chemical Engineering”: The first two keywords are more reflective of the current context of the times. It broke out in 2018 and continues to this day. In the Notice of the General Office of the Ministry of Education on the Announcement of the First Batch of Research and Practice Projects of “New Engineering” dated March 15, 2018, it was emphasized that all relevant units should actively implement the construction of “New Engineering”, and push forward the implementation of the projects with the spirit of striving for the best and waiting for the best. On September 17 of the same year, the Ministry of Education, the Ministry of Industry and Information Technology, and the Chinese Academy of Engineering jointly pointed out the need to accelerate the construction and development of new engineering disciplines, explore the formation of an engineering education system with Chinese characteristics and world standards, and promote China’s development from a large engineering education country to a strong one. With the development of the internet, the cooperation between the new engineering disciplines with the internet and artificial intelligence as the core and universities has become more and more diverse, which has greatly improved the possibility of innovation and entrepreneurship for college students. The reform of innovation and entrepreneurship education involves a wide range of majors, among which the research on food majors and chemical majors is relatively rich. In this background of new engineering, cultivating high-quality composite new engineering talents with strong practical ability, innovation ability and international competitiveness has become the main goal, so the educational reform of chemical engineering majors becomes crucial.
- (3) “Higher education institutions”, “integration of industry and education”, “collaborative education”: exploding in 2019 and continuing to date. The “integration of industry and education” is a new way of cooperation between industry and universities and is an urgent requirement to promote the structural reform of human resources supply side. This slogan has been put into practice in our country for some time. However, the ecological environment for the implementation of “industry-education integration” in China’s institutions of higher education is not ideal, and there are problems such as confusion in school-enterprise management mode, difficulties in transforming students’ apprenticeship status and low teachers’ levels. If we want to cultivate truly innovative talents for the country, the relevant departments need to rectify these problems (Lin *et al.*, 2021).
- (4) “Rural revitalization”, “labor education”, “food professions”: exploding in 2019 and continuing to date. In 2017, China proposed a rural revitalization strategy. Starting in 2019, many scholars began to conduct research on innovation and entrepreneurship reform in universities in this context. Lv (2022) proposed that labor education should be included in the overall plan for cultivating socialist builders and that giving full play to the educating function of labor education is an inevitable choice for the development of the times and inevitable choice for activating the driving force of rural revitalization. Meng *et al.* (2020) pointed out that the food major should join hands with the agriculture and processing industries, give full play to its professional advantages in the rural revitalization strategy, and provide talent supply and technical support for solving the “three rural” problems.

## 6. Conclusion and outlook

An era of mass entrepreneurship and innovation is coming to us, the new era urgently needs to cultivate a large number of innovative and entrepreneurial talents. This paper takes “innovation and entrepreneurship education” as a research object, collects relevant literature from core journals in CNKI and WOS, and uses *Citespace* to make keyword co-occurrence graphs, topic clustering graphs, time-series graphs, and Excel to construct trend and frequency distribution graphs. It also describes and illustrates the research history of dual innovation education at home and abroad from 2000 to the present and analyzes the current research field and trends at home and abroad.

### 6.1 Analysis of the problem

In the 1990s, when Chinese innovation and entrepreneurship education was just in its infancy, a mature system had already been developed abroad. In recent years, China has continuously introduced a series of policies to support the development of innovation and entrepreneurship education, but there are still widespread problems of immaturity in the system of specialization and innovation integration, such as low level of university collaboration, needy competence of teachers, etc. which lay behind by foreign countries.

*6.1.1 Insufficient reform efforts and putting the cart before the horse in terms of direction.* Cultivating entrepreneurial consciousness and entrepreneurship among college students is the essence of entrepreneurship education. The United States and other developed countries have penetrated students' innovative and entrepreneurial consciousness into every aspect of college education, while domestic research on this field is still in the primary stage, putting more energy on the employment of graduates without implicitly cultivating students' innovative and entrepreneurial spirit. To a certain extent, China's innovation and entrepreneurship education has a utilitarian and simplistic tendency, which leads to the lagging innovation concept and weak entrepreneurial consciousness of students in domestic colleges and universities. At present, some domestic universities offer innovation and entrepreneurship courses, but most of these courses exist in the form of elective courses. In addition, the existing innovation and entrepreneurship courses are too isolated from each other lacking horizontal connection, which makes it difficult to form an effective curriculum system and thus difficult to form the synergy of talent cultivation. This is mainly due to the fact that most universities are lagging behind in concept and do not realize that innovation and entrepreneurship education is a systematic project.

*6.1.2 Uneven distribution of resources and insufficient strength of teachers.* Foreign countries attach great importance to the training and cultivation of teachers, but China's domestic awareness in this area is relatively weak. There is an extreme shortage of professional dual-innovation teachers, which is not only reflected in underdeveloped and remote areas, but even in first-tier cities such as Beijing, Shanghai, Guangzhou and Shenzhen. The faculty of entrepreneurship education in developed countries is mainly composed of government officials, successful entrepreneurs and senior experts, who have rich experience in entrepreneurship. However, most of them are composed of teachers and counselors in colleges and universities. Although they are more knowledgeable, most of them have not received systematic innovation and entrepreneurship education and lack innovation and entrepreneurship experience and practical guidance ability, and they are more inclined to theoretical lectures. Since most domestic colleges and universities obviously lack training and support for teachers of innovation and entrepreneurship education, this leads to the lack of full-time professional teachers of innovation and entrepreneurship education. As for entrepreneurs, they lack the motivation to engage in innovation and entrepreneurship education. Due to the lack of organic linkage between schools and enterprises, quite a number of entrepreneurs simply equate college students' innovation and entrepreneurship with



college students' employment internships, which makes college students' innovation and entrepreneurship education eventually become a formality.

*6.1.3 Backward curriculum design and poor integration of specialization and creativity.* Although China's current research in this field focuses partly on the study of specialization and integration, the practical effect is still unsatisfactory. Curriculum design in colleges and universities generally favors fixed thinking, and most college students receive the same education, lacking diversified development and slow updating speed. And at present, the strength of specialization and integration in our universities is far from enough. Only some content about innovation and entrepreneurship is superficially involved in some ideology and politics courses, without organically integrating various professional disciplines with innovation and entrepreneurship education to achieve synergistic effects.

*6.1.4 Low degree of synergy and vague subject orientation.* The phenomenon of low degree of synergy in this field in China is not only reflected in the large level of government, universities, society, and students, but also in the small level between various departments within universities. From the large level, the above four main bodies all participate in the reform of innovation and entrepreneurship education in their independent identities and different ways, and there are fewer communication links between different bodies and closed information feedback channels, which do not maximize their proper value and the breakthroughs that collaboration can bring. At a small level, the internal framework of the universities has not been redesigned according to the needs of innovation and entrepreneurship education. Some universities have set up special innovation and entrepreneurship centers and laboratories, etc., but they are still only independent individuals, and other departments do not cooperate with them enough, so the effect is not satisfactory.

## 6.2 Solutions

Looking forward to the research and development of innovation and entrepreneurship education, China should learn from the excellent experience of foreign countries and continuously explore the new model of innovation and entrepreneurship education in the post-epidemic age with combination of its own situation.

*6.2.1 Establishing an ecosystem of positive interaction in innovation and entrepreneurship education.* The ecosystem of dual innovation education mainly includes three main bodies: universities, government and society. First of all, it is necessary to continuously improve the teaching system within the university, improve the faculty strength and integrate various resources to build a professional innovation and entrepreneurship education faculty. Secondly, the government should provide policy guarantee for the development of innovation and entrepreneurship education, encourage universities to carry out innovation and entrepreneurship activities, provide certain financial support for the participants of innovation and entrepreneurship competitions and give corresponding rewards to the winners, to ensure the driving force of dual-innovation education (Wang *et al.*, 2023). Finally, the society should create a good atmosphere for innovation and entrepreneurship, unite with schools to carry out a series of dual-innovation activities, provide students with some off-campus practice opportunities, and promote the positive development of the innovation and entrepreneurship ecosystem.

*6.2.2 Reconstructing the curriculum system of colleges and universities with deep integration of specialization and innovation.* According to the analysis of time series of keywords, China started to increase the research on "integration of specialization and innovation" in 2020 and has achieved certain results, but the main measures taken by universities are only the simple "integration" of professional courses and innovation and entrepreneurship courses, instead of "integration". Innovative entrepreneurship education has the function of lifelong and humanities education, as well as scientific and professional education (Wang, 2021a, b). To promote the deep integration of professional and creative education, it is necessary not

only to improve the content of the curriculum but also to carry out a comprehensive reform of the teaching system, the teaching staff and the curriculum. Firstly, we can integrate innovative thinking into the teaching of professional courses and increase classroom interaction; secondly, we can promote cross-disciplinary learning, such as adding a double degree of “admission major” + “innovation and entrepreneurship major”; thirdly, we can increase practical opportunities for students and turn innovation and entrepreneurship theories into practical actions to increase students’ interest. For example, the University of Georgia CCURO in the USA awards 30 summer research scholars each year to support undergraduate students who are interested in the training of scientific research during the summer (Long *et al.*, 2021). Finally, the focus should be on a deep reform of the chemical and food majors to cultivate relevant talents to meet the needs of China’s new engineering background and rural revitalization strategy.

#### *6.2.3 Strengthen the professional training for teachers of innovation and entrepreneurship.*

In the reform of dual-innovation education, teachers are one of the key factors that can have a direct impact on students’ perception of innovation and entrepreneurship. Foreign universities also attach great importance to this factor of teachers and have invested a lot of money and human resources with obvious results. All major universities in China should cultivate a team of innovative and entrepreneurial teachers and regularly provide them with professional training to improve their own innovative and entrepreneurial abilities and teaching levels, thus providing quality teaching support for the cultivation of students’ innovative and entrepreneurial spirit and abilities. We can also invite well-known entrepreneurs from the alumni or people who are successfully starting their businesses to give lectures on campus to share students with their own experiences, stimulate students’ enthusiasm for entrepreneurship and raise their initiative awareness. At the same time, there is a need to recognize that innovational centrality presents a unique set of complexities for institutional stakeholders as they provide students with the tools they need to build a regional creative economy (Selznick, 2019).

#### *6.2.4 Strengthening cooperation among scholars and promoting collaborative innovation in universities.*

The government and universities should explore and cultivate potential authors, and attract more scholars to invest in the research of dual innovation education through policy incentives and financial support. However, the educational authorities should implement a “one-school-at-a-time” policy other than a one-size-fits-all approach, giving universities the freedom to carry out entrepreneurship education and research, aiming to better promote local economic and social development and form a various and extensive innovation and entrepreneurship education system (Zhang and Zeng, 2021); the educational authorities should play the role of core authors, form a group of core authors, and gradually build a professional research team, with anticipation to produce higher-level scientific research results; they also should build a good academic exchange platform, and promote communication and cooperation among scholars, especially between core authors to promote the development of innovation and entrepreneurship education research. At the same time, colleges and universities should build a collaborative innovation mechanism, a concept that still originated in the US Synergy does not only exist between colleges and enterprises but is also embedded in the mutual influence between colleges and universities. Therefore, building a collaborative mechanism among colleges and universities can break communication barriers, make the most efficient use of innovation and entrepreneurship resources, and put the reform of dual-innovation education on the “accelerator belt”.

### *6.3 Implications*

#### *6.3.1 Implications for researchers’ policymakers and practitioners.*

This paper compares the relevant literature in the field of innovation and entrepreneurship education at home and abroad based on the bibliometric method, draws a knowledge map of the field, shows

the development pulse of the field and detects the frontier development of the field. Therefore, this study has the following main implications for relevant researchers: it can help researchers who are new to the field of innovation and entrepreneurship education to get a quick overview of the development of the field; it can help researchers to understand the current development of the field and the problems that exist, and carry out targeted research; it can help researchers to obtain the latest information and research trends in the field, improve academic efficiency and seize cutting-edge research topics.

*6.3.2 Implications for policymakers.* This paper provides a comparative analysis of the development of innovation and entrepreneurship education at home and abroad, and offers comments and suggestions. This study can help policymakers to recognize the real development of innovation and entrepreneurship education in China and to understand the efficiency and quality of current reform policies, so that relevant policies can be formulated that are more in line with China's national conditions and more practical.

*6.3.3 Implications for colleges and universities.* In the in-depth analysis of the knowledge graph, this paper found many problems existing in the innovation and entrepreneurship education in domestic universities, and put forward relevant suggestions. This study can effectively help universities to better position their own role in the reform of innovation and entrepreneurship education, better focus on the national development and students' own needs, improve the cultivation mode and enhance the efficiency of talent cultivation.

#### *6.4 Limitations and prospects*

This paper presents a more systematic bibliometric analysis of the research field of innovation and entrepreneurship education at home and abroad, further analyses the current problems in China through comparison, and proposes corresponding solutions based on China's education system and development direction, with certain practical significance. However, the study still has some limitations.

Firstly, the research method of this paper has certain shortcomings, as it analyses the development of a certain field through the measurement of literature, which is weakly integrated with practice; secondly, the searched literature of this paper is not comprehensive enough, and only 9,960 papers published in core journals are selected for the requirement of paper quality; finally, as the purpose of this paper is to make suggestions for the reform of innovation and entrepreneurship education in China, it does not analyze the shortcomings of foreign innovation and entrepreneurship education system in more depth, and the contrast presented is not strong enough.

In the subsequent study, the limitations of this paper should be broken down to make the study more persuasive. Firstly, it can add the combing of national policies to objectively demonstrate the concept and focus of China's innovation and entrepreneurship education reform; secondly, it can add the content of questionnaire surveys, which include university teachers and students, to gain an in-depth understanding of the ideas of participants and beneficiaries in the education reform; finally, it can expand the research database as well as the time scope, and add comparative analysis from home and abroad to improve the academic nature of the article.

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### Further reading

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